

Unsupervised Role Discovery from Professional Counter-Strike 2 Demo Data

Background

This project analyzes professional Counter-Strike 2 demo files to discover player roles using unsupervised machine learning. Instead of training a supervised model on predefined labels, the system attempts to determine whether recognizable player archetypes emerge naturally from gameplay behavior alone.

The pipeline uses raw .dem replay files (demos) from top-tier professional tournaments, engineers behavioral and positional features, applies multiple clustering algorithms, and evaluates the resulting role structures using both traditional clustering metrics and a custom subsampling stability framework.

Professional Counter-Strike provides an interesting environment for this type of analysis because player roles are partially structured but are not rigidly defined. While some roles, such as dedicated AWPers, are relatively easy to recognize, others depend heavily on positioning, utility usage, timing, or team coordination. This makes role discovery an interesting and challenging machine learning problem with substantial ambiguity and noise.

The project currently uses professional match demos from four S-tier events: IEM Atlanta 2026, PGL Astana 2026, CS Asia Championship 2026, and the entire IEM Cologne Major. It is unlikely that new data outside of these four tournaments would be added. The roster lock for the IEM Cologne Major reduces roster turnover and makes player profiles more comparable across different games and events. The dataset consists of 463 demos and covers 214 professional players.

The project was implemented entirely in Python using awpy [1] for demo parsing, Polars and pandas for data processing, scikit-learn [2] for clustering and evaluation, HDBSCAN [3] for density-based clustering, and Matplotlib for visualization generation.

Problem Statement

Professional Counter-Strike teams are often described through player roles such as entry fraggers, lurkers, anchors, support players, AWPers, and in-game leaders. These roles are useful for analysis, but they are neither rigid nor mutually exclusive.

A player's responsibilities can shift by side, map, economy state, opponent, roster context, or team system, making role classification inherently noisy. This project uses expert-curated role annotations from NER0cs's Positions Database [4] as an external reference point, while still treating role discovery as an unsupervised learning problem. The model is not trained to reproduce predefined labels. The labels are used only after clustering to evaluate whether clusters align with established classifications.

The core question is:

Can professional CS2 player roles be discovered from behavioral and positional gameplay statistics, and how closely do those discovered clusters align with expert-curated role annotations?

Pipeline

The project uses an automated data pipeline that transforms raw professional Counter-Strike 2 match demos into clustered player role profiles.

The pipeline begins with automated extraction and deduplication of professional match demos downloaded manually from HLTV.org [5]. A custom extraction script scans recent downloads, extracts valid .dem replay files, removes duplicates, and moves new demos into the project dataset directory.

Once extracted, demos are processed through a multiprocessing-based parser built on top of the awpy library. Awpy allows the parser to extract a wide range of gameplay information including combat events, utility usage, opening duels, trade interactions, movement behavior, and positional relationships between players.

A large feature engineering stage transforms this raw data into behavioral statistics describing how players fight, move, coordinate with teammates, and use utility. Count-based features are converted to per-round rates, positional measures are averaged over replay observations, and selected features include across-demo standard deviations to represent behavioral consistency. All model features are

standardized before clustering. The final dataset stores one row per player per side (CT and T), allowing side-specific behaviors to be analyzed independently.

The pipeline then evaluates multiple unsupervised clustering algorithms including KMeans, Gaussian Mixture Models (GMM), and HDBSCAN. Each clustering configuration is evaluated using silhouette score, Davies-Bouldin index, and a custom subsampling stability analysis based on Adjusted Rand Index (ARI). The stability analysis repeatedly resamples the dataset and recomputes clusters to measure how consistently discovered structures reappear. External validation was performed using expert-curated role labels from a database maintained by Harry “NER0cs” Richards, a Counter-Strike analyst and writer for hltv.org. These labels were used as reference annotations to evaluate whether the unsupervised clusters aligned with established professional role classifications. Because CS2 roles can vary by side, map, team system, and roster context, the labels are treated as expert reference labels rather than absolute ground truth.

PCA scatter plots, radar charts, stability bar charts, silhouette-versus-k plots, and z-score heatmaps are created to assist with analysis. These visualizations make it possible to identify separable role structures, detect outlier players, and understand which gameplay characteristics define each cluster.

Role Definitions

The external role labels describe each player’s primary responsibility rather than an action performed in every round. Professional players often perform multiple responsibilities, so these categories should be interpreted as dominant behavioral archetypes rather than strict classifications.

An AWPPer primarily uses the AWP, a long-range sniper rifle typically deployed from powerful map positions to control key sightlines, secure opening picks, and create opportunities for the team. An IGL, or in-game leader, directs team strategy and makes mid-round decisions. Because IGL is a leadership designation rather than a mechanical role, it can overlap with any of the other categories.

On the CT-side, an Anchor is responsible for defending a bombsite, often remaining in position alone when teammates rotate. On the other hand, a Rotator moves between areas in response to information and reinforces teammates when pressure develops elsewhere.

On the T-side, a Spacemaker initiates or supports contested map control and frequently participates in engagements. A Lurker operates separately from the main attacking group to control rotations, maintain map pressure, and exploit gaps in the defense.

Feature Engineering

Feature engineering was a major part of the project. Since player roles in professional Counter-Strike are not directly labeled, clustering quality depended heavily on building features that captured meaningful differences in playstyle rather than just raw performance. Instead of relying only on standard statistics such as ADR or kills per round, the dataset includes metrics intended to describe how players take space, interact with teammates, use utility, and engage opponents. All features are normalized by rounds played, and both averages and standard deviations are recorded to capture behavioral consistency as well as central tendency.

Combat features include kills per round, survival rate, damage per round, damage taken per round, multi-kill rate, and rifle and AWP kill shares. ADR and damage differential are retained as descriptive summaries but are excluded from clustering to avoid redundant representations.

Opening duel metrics measure early-round aggression and entry tendencies through opening kill rate, opening death rate, opening duel success rate, and first contact rate. Players who consistently make first contact with the enemy show a distinct behavioral signature from players who may play more conservatively before engaging.

Trading metrics capture how players operate within team structure. A trade kill is defined as a kill that occurs within five seconds of a teammate's death against the player who made that kill. Trade kill rate, death traded rate, and trade participation together help distinguish coordinated support players from lurkers and independent fraggers who operate in isolation.

Utility metrics record grenade usage rates, flash assists, utility damage, smoke usage, and incendiary usage per round. These features are particularly important for identifying players whose primary contribution comes from enabling teammates rather than direct combat output, which helps characterize the utility-oriented, IGL cluster on the T-side.

Finally, positional metrics are computed directly from replay tick data and describe where players spend their time relative to teammates and opponents. These include average distance to enemies, distance to the team centroid, nearest teammate distance, movement distance per round, time spent near enemies, and stationary behavior rate. Positional features were the dominant drivers of cluster separation on the CT-side, where the distinction between players is expressed more through spacing than through combat statistics.

Clustering Methodology

KMeans, Gaussian Mixture Models (GMM), and HDBSCAN were each included because they capture different types of structure within the data. KMeans partitions players into compact clusters based on centroid distance, while GMM models clusters probabilistically, allowing softer boundaries between overlapping playstyles. HDBSCAN was included to test whether the data contained density-based groups or outlier profiles without forcing every player into a cluster.

Before clustering, all numeric features were standardized using StandardScaler so that features with larger numeric ranges would not dominate the clustering process. A Principal Component Analysis (PCA) was also computed for visualization purposes, allowing high-dimensional player profiles to be projected into two dimensions for interpretation.

Models were ranked by a composite score combining silhouette score (40%), an inverted Davies-Bouldin index (20%), and subsampling stability (40%), where the stability term was penalized by its own standard deviation to discount models whose cluster assignments varied substantially across resamples. Silhouette score measures how well-separated clusters are, while Davies-Bouldin index measures cluster compactness and overlap. However, these metrics alone can produce misleading results in high-dimensional spaces, especially in noisy behavioral data.

To address this, the project implements a subsampling stability analysis based on ARI. For each clustering configuration, 80% of the player-side profiles are repeatedly resampled without replacement and reclustered. The resulting cluster assignments are then compared against the original clustering solution using ARI, measuring how consistently the discovered structures reappear.

Results and Analysis

CT Side

The CT-side results are best represented by the kmeans_k3 solution. The model produces three interpretable clusters corresponding broadly to Rotators, Anchors, and AWPers. It achieves a silhouette score of 0.166, a Davies-Bouldin index of 2.035, and a composite model-selection score of 0.523. Its mean subsampling stability ARI is 0.992 ± 0.014 across repeated 80% subsamples without replacement. This high stability indicates that the CT cluster structure is reproduced consistently when the player pool changes. Of the 157 eligible CT-side player profiles, 123 were matched to the external CT-specific role labels. Against these matched labels, kmeans_k3 achieves an ARI of 0.612 and an overall matched-sample purity of 78.9%. The three clusters have the following interpretations.

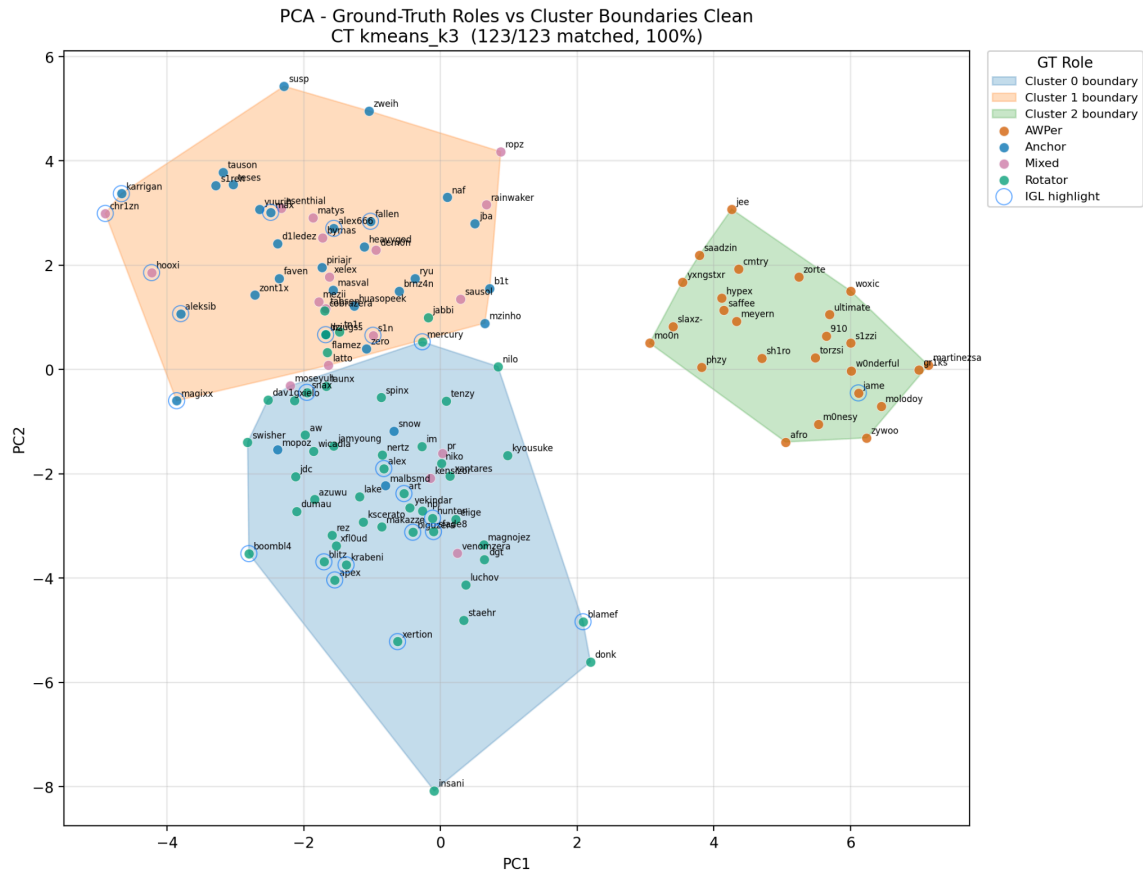


Figure 1. PCA projection of CT-side kmeans_k3 clusters and external roles for 123 matched players.

Cluster 0 - Rotators, 86.5% purity: This cluster contains 65 players, of whom 52 were matched to the external labels. Forty-five of the matched players are labeled as Rotators. It is the cleanest non-AWPPer CT cluster and suggests that Rotators have a relatively consistent behavioral profile. The model-level feature-importance results show that distance to the team centroid, relative team-centroid distance, nearest-teammate distance, distance to enemies, time near enemies, and first-contact behavior all contribute to the CT partition. These variables are consistent with a role that involves repositioning, reinforcing teammates, and responding to pressure across the map according to round state and information.

Cluster 1 - Anchors, 58.7% purity: This cluster contains 61 players, with 46 matched to the external labels. Twenty-seven of those matched players were labeled as Anchors. The lower purity indicates that Anchor behavior may be less separable than Rotator behavior. This is plausible because anchoring is highly dependent on map, bombsite, opponent, and team system. Some Anchors hold isolated positions and see little early action, while others are frequently pressured or play more active defensive

setups. The cluster therefore captures an Anchor-like behavioral group but also includes CT riflers whose aggregate statistics overlap with those of Anchors.

Cluster 2 - AWPers, 100.0% purity: This cluster contains 31 players, of whom 25 were matched to the external CT labels. All 25 matched players are labeled as AWPers. The cluster is defined most clearly by weapon-profile variables, particularly AWP kill share and rifle kill share, although positional and contact-related variables also contribute. The perfect matched-sample purity confirms that the AWP role produces the clearest signal on the CT-side.

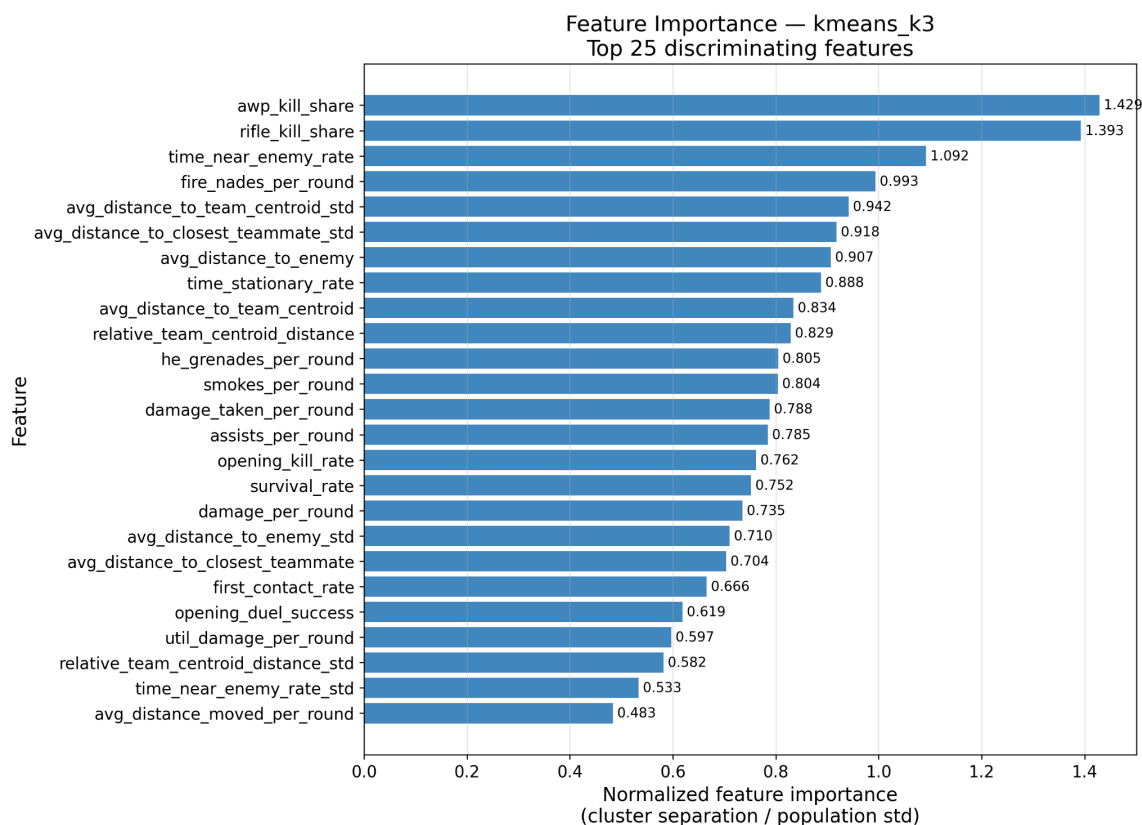


Figure 2. Top 25 features distinguishing the CT-side kmeans_k3 clusters.

The model does not recover a distinct CT-side IGL cluster. Its ARI against the binary IGL labels is -0.023, and its balanced accuracy is 0.500, indicating no meaningful separation beyond the majority-class baseline. CT-side IGLs are instead distributed among the Rotator, Anchor, and AWPPer clusters. This is consistent with IGL being a leadership responsibility rather than a single mechanical role. An IGL may anchor, rotate, or AWP depending on the team and map. In addition, another experienced player may contribute to CT-side calling, allowing the primary IGL to focus more of their preparation on the T-side.

Moving from three to four KMeans clusters does not improve the CT interpretation. Although `kmeans_k4` retains the broad role structure, its stability falls to 0.828 ± 0.129 , its composite score falls to 0.398, and its external ARI falls to 0.462. The additional cluster therefore subdivides the rifler population without producing a stronger or more stable role interpretation.

Overall, the CT results show that aggregated demo statistics recover a highly stable AWPer cluster and a meaningful but imperfect Rotator/Anchor split. Positional relationships and contact behavior are particularly important for distinguishing the two rifler groups, while IGL status does not emerge as an independent CT-side category.

T Side

The T-side clustering results are best summarized by the `kmeans_k4` solution. The model produces four reasonably interpretable clusters while avoiding the small, fragmented groups found in higher-k solutions. It achieves a silhouette score of 0.138, a Davies-Bouldin index of 2.018, and a composite model-selection score of 0.427. Its mean stability ARI is 0.835 ± 0.072 across repeated 80% subsamples without replacement. This stability is lower than the CT-side result, suggesting that T-side roles are less rigidly separated in the feature space.

Against the T-specific ground-truth labels, the model reaches an ARI of 0.456 and an overall matched-label purity of 78.9%. Of the 162 eligible T-side players, 123 could be matched to the external role labels used for evaluation. The PCA visualization shows that the AWPer cluster is distinct, as expected, while the three rifler clusters overlap more substantially. Four clusters emerge:

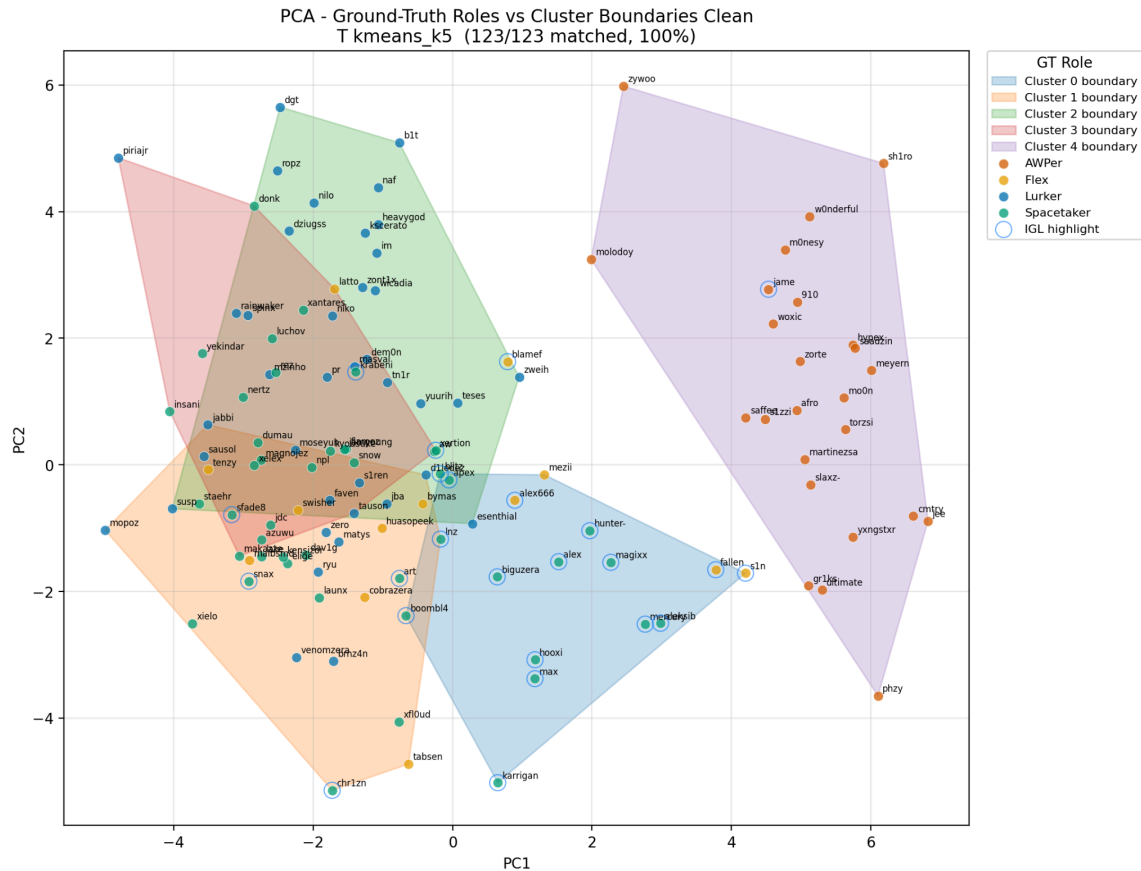


Figure 3. PCA projection of T-side kmeans_k4 clusters and external roles for 123 matched players.

Cluster 0: Aggressive Spacetakers, 61.7% purity. This is the largest cluster, containing 67 players, of whom 47 have matched T-side labels. Twenty-nine of those matched players are labeled as Spacetakers. The cluster is characterized by high first-contact involvement, opening kills, opening deaths, damage taken, and time spent near enemies. Players in this group also have lower survival rates and shorter average distances from opponents. These features describe aggressive riflers who frequently contest defended areas and create space for their teammates. The moderate purity indicates that this behavioral profile is not exclusive to players formally labeled as Spacetakers, but the cluster still captures a recognizable high-contact role.

Cluster 1: Utility-Oriented Spacetakers and IGLs, 76.2% purity. This cluster contains 28 players, with 21 matched to the external T-side labels. Sixteen of the matched players are labeled as Spacetakers. Compared with Cluster 0, these players have lower damage, kills per round, multi-kill frequency, and opening success, but use substantially more utility. The cluster therefore represents a more supportive or enabling form of space-taking rather than a purely aggressive entry profile.

This cluster also contains a strong concentration of in-game leaders (IGLs). Seventeen of its 21 matched players are labeled as IGLs in the binary IGL evaluation. This produces an IGL purity of 81.0% within the cluster. However, the overall ARI against the IGL labels is only 0.147. The model therefore identifies one IGL-enriched behavioral group, but it does not recover IGL status as a clean global division across the entire player population. This is plausible because leadership is not directly observable through demo statistics and because IGLs can perform several different roles. Furthermore, the model does not have access to team communications or strategic calls, so it can detect only the indirect behavioral effects associated with leadership.

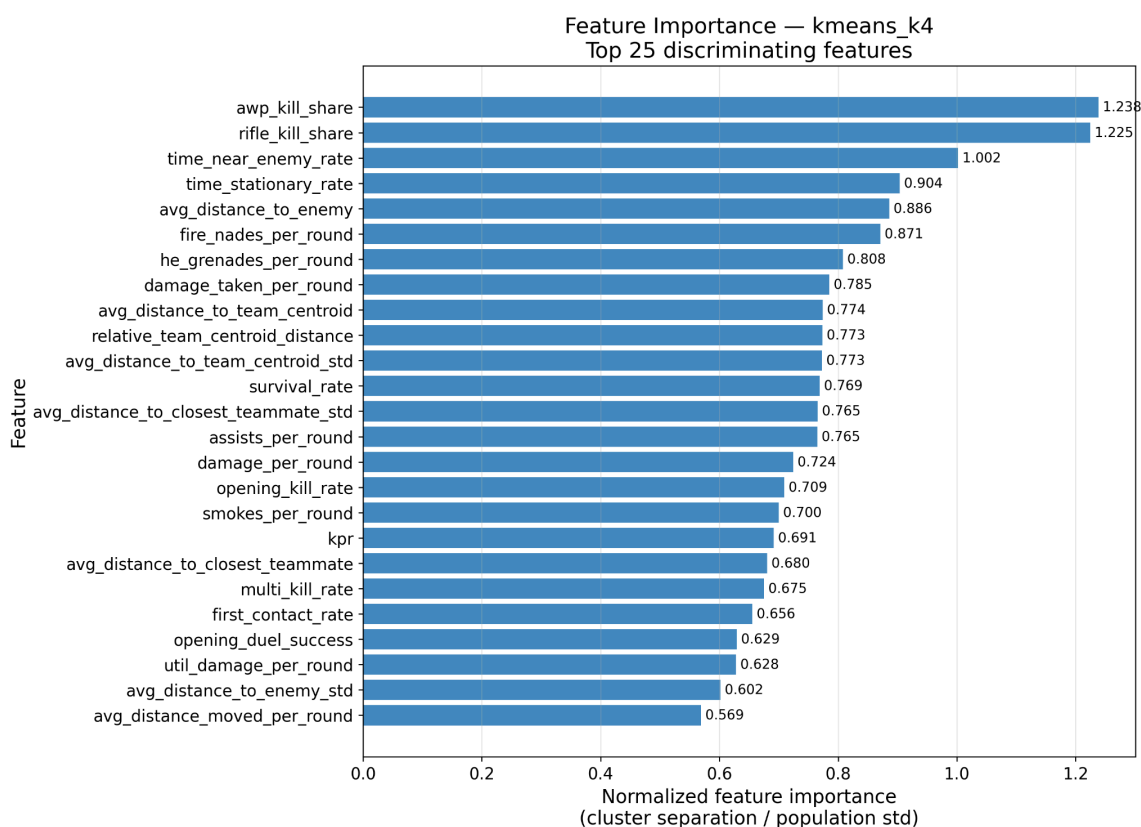


Figure 4. Top 25 features distinguishing the T-side kmeans_k4 clusters.

Cluster 2: AWPers, 100.0% purity. This cluster contains 32 players, of whom 25 have matched T-side role labels. All 25 are labeled as AWPers. The cluster has by far the highest AWP kill share and the lowest rifle kill share. Its players also tend to take less damage, die less frequently in opening engagements, and spend less time close to enemies. These characteristics reflect the longer-range and more selective engagement pattern associated with primary AWPers. As on the CT-side, weapon usage creates the clearest signal in the dataset, making the AWPPer the most consistently recovered role.

Cluster 3: Lurkers, 90.0% purity. This cluster contains 35 players, with 30 matched to the T-side ground truth. Twenty-seven of those players are labeled as Lurkers. The cluster is distinguished primarily by spatial separation. Players have greater average distance from their teammates and the team centroid, as well as greater variability in these positional measures across demos. This profile is consistent with lurkers, who frequently operate away from the main attacking group to control rotations, apply pressure, and exploit openings elsewhere on the map. The high purity indicates that positional isolation is a strong behavioral signal for identifying the Lurker role.

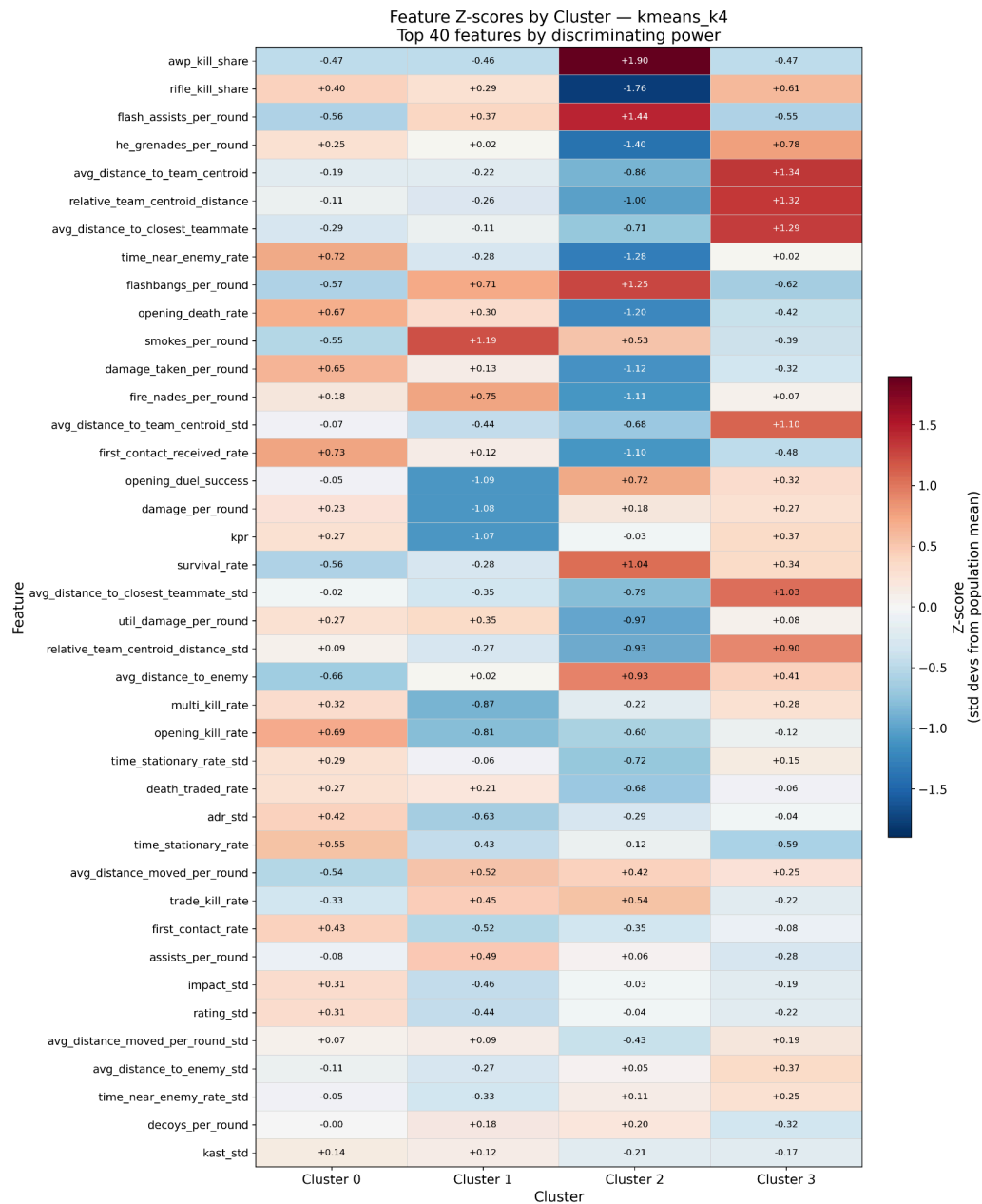


Figure 5. Standardized feature profiles for the four T-side kmeans_k4 clusters.

Overall, the T-side results indicate that meaningful role structure is present, but it is less compact and less stable than on the CT-side. The model strongly recovers AWPers and Lurkers, while dividing Spacetakers into an aggressive high-contact group and a lower-output, utility-oriented group that is enriched with IGLs. The lower stability and greater overlap between the rifler clusters are consistent with the fluidity of T-side play, where responsibilities can change according to map, strategy, spawn, and round context. For these reasons, kmeans_k4 provides the best balance of validation performance, stability, external agreement, and interpretability for the T-side.

Cross Model Observations

Across all solutions, AWPers form the cleanest clusters on both sides. AWP kill share is among the strongest distinguishing features in both models, showing that weapon specialization creates a clearer signal than most rifler behaviors.

KMeans produces the strongest non-HDBSCAN solutions on both sides. CT kmeans_k3 achieves a stability ARI of 0.992 and a composite score of 0.523, while T kmeans_k4 achieves a stability ARI of 0.835 and a composite score of 0.427. GMM is generally less stable when compared to KMeans solutions. With the str 0.906 for CT gmm_k3 and 0.717 for T gmm_k4. Their larger variation across subsamples makes them less reliable than the selected KMeans solutions.

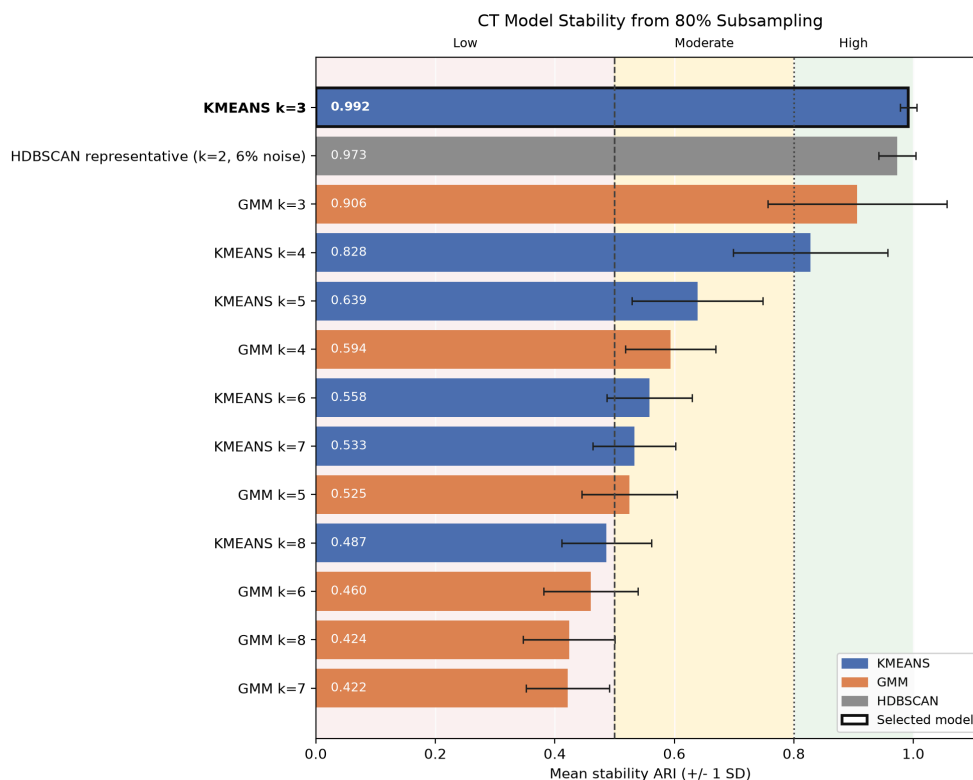


Figure 6. CT model stability under repeated 80% subsampling.

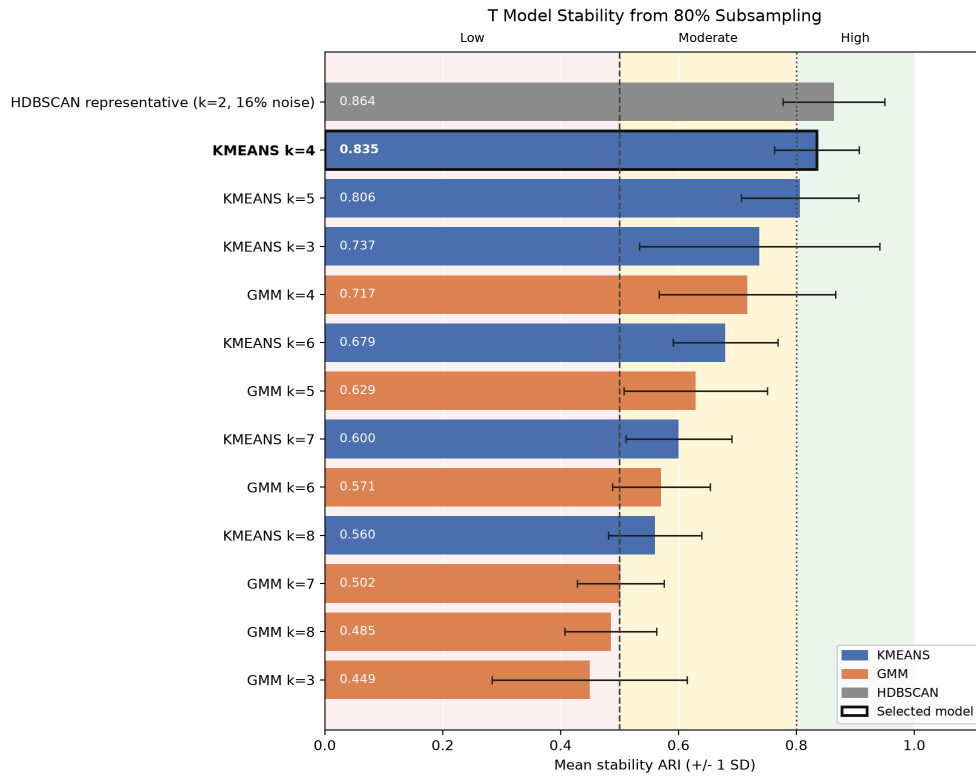


Figure 7. T model stability under repeated 80% subsampling.

HDBSCAN produced highly stable but overly broad two-cluster solutions that primarily separate the strongest weapon-profile structure from the remaining players. The leading CT configuration has a stability ARI of approximately 0.973 with 6.4% of players labeled as noise, while the leading T configuration has a stability ARI of approximately 0.864 with 16.1% labeled as noise. These solutions score well on internal metrics but do not recover the more detailed rifler roles sought by the project. This demonstrates that strong internal stability does not necessarily produce the most useful role interpretation.

Ablation Analysis

An ablation analysis was conducted to determine which feature groups drive the CT- and T-side cluster structures. Every CT ablation was evaluated using three clusters, while every T ablation was evaluated using four clusters. Holding the number of clusters constant makes it possible to compare feature representations without allowing changes in k to create an additional source of variation.

The experiments included removal-based ablations, in which one feature category was excluded from the full model, and category-only ablations, in which clustering was performed using a single feature group. Results were evaluated using internal separation, subsampling stability, composite score, external-label ARI, and matched-sample purity.

CT-Side Ablations

The full CT model achieves a stability ARI of 0.992, an external-label ARI of 0.612, a matched-sample purity of 78.9%, and a composite score of 0.523. Removing trading and teamwork features produces nearly identical results, with an ARI of 0.612, purity of 78.9%, and a composite score of 0.524. Removing utility features also preserves the same ARI and purity, with a slightly lower composite score of 0.514. These results indicate that trading and utility variables help describe the clusters but are not required to recover the main CT structure.

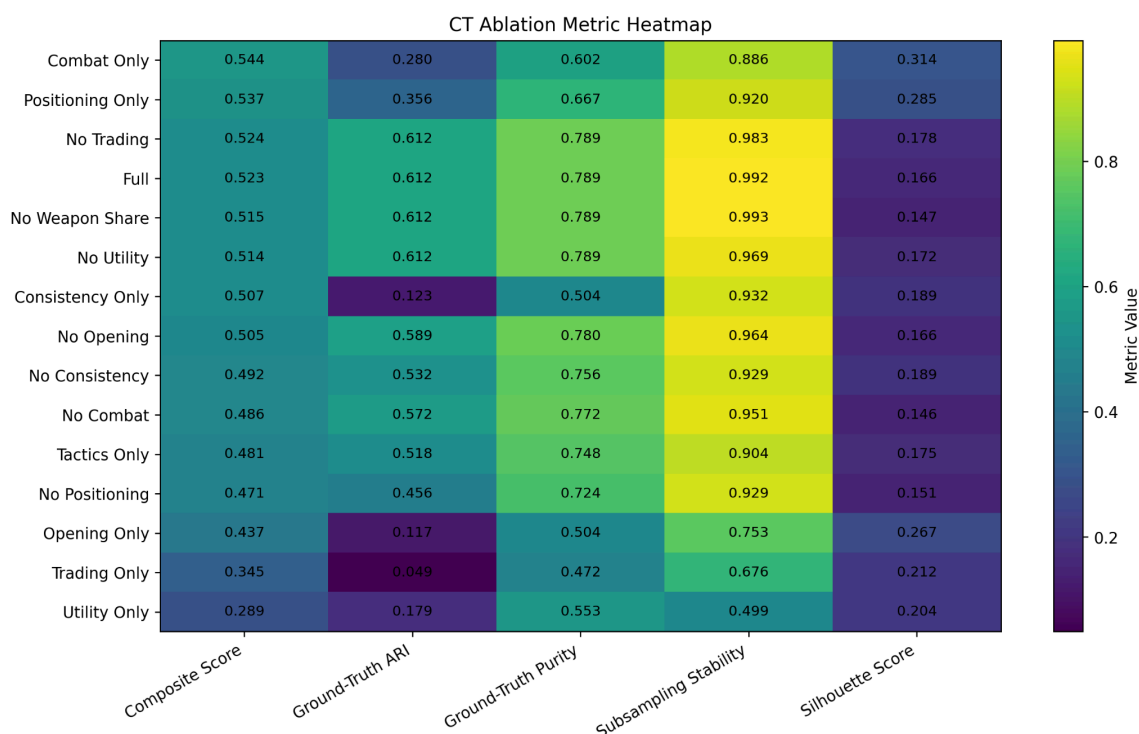


Figure 8. CT-side ablation results with $k = 3$ held constant.

Removing weapon-share features also retains an external-label ARI of 0.612 and a purity of 78.9%. This shows that overall alignment with the external CT roles remains strong even without direct AWP and rifle kill-share variables. Other correlated features may preserve part of the same role structure, although weapon-share variables make the AWP interpretation more direct.

The largest removal-based decline occurs when positioning and movement features are excluded. External ARI falls from 0.612 to 0.456, purity falls from 78.9% to 72.4%, and the composite score falls from 0.523 to 0.471. The clusters remain reasonably stable, but their agreement with the CT-specific role labels becomes substantially weaker. This provides the strongest evidence that the Rotator/Anchor distinction depends on spatial behavior.

This result is consistent with the difference between Rotators and Anchors. Rotators are expected to reposition in response to mid-round information, while Anchors more often hold fixed defensive responsibilities within a site. Because CT-side actions are typically reactive to T-side pressure and executes, spatial behavior is more informative than raw combat or utility statistics for distinguishing the CT rifler roles.

The category-only experiments reinforce this conclusion. Positioning and movement features alone produce a stability score of 0.920 and an ARI of 0.356. Combat features alone form compact internal clusters, achieving the highest CT silhouette score of 0.314, but their ARI is only 0.280. Trading and teamwork features alone produce an ARI of 0.049, while utility features alone produce an ARI of 0.179.

Overall, the CT ablations indicate that the complete role structure depends primarily on positional and movement information combined with weapon and combat signals. Utility and trading features provide additional behavioral detail but have limited influence on the final external role alignment. AWPers remain identifiable through several correlated feature groups, while the Rotator/Anchor rifler split is most sensitive to the removal of spatial information.

T-Side Ablations

The full T-side model achieves a stability ARI of 0.835, an external-label ARI of 0.456, a matched-sample purity of 78.9%, and a composite score of 0.427. Unlike on the CT-side, no individual feature group reproduces the complete T-side interpretation. Different feature categories capture different aspects of the Spacetaker, Lurker, AWPer, and IGL-enriched profiles.

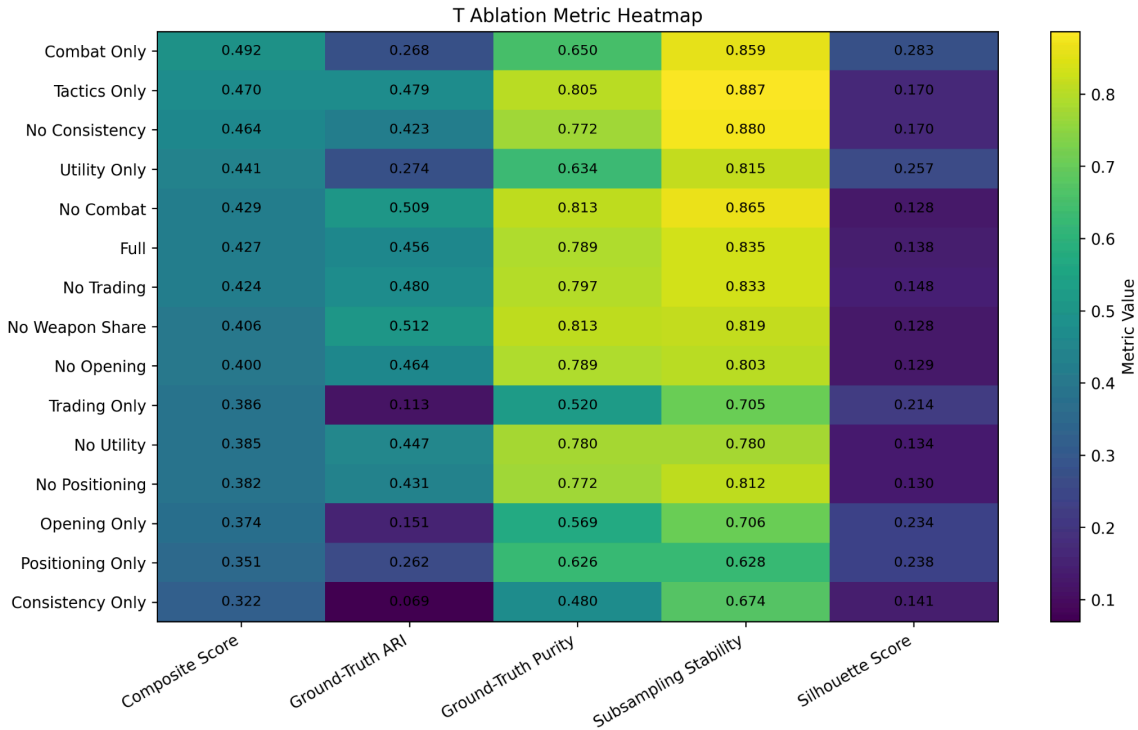


Figure 9. T-side ablation results with $k = 4$ held constant.

The tactics-only model combines contact, teamwork, utility, positioning, and movement features while excluding direct combat, weapon-share, and consistency variables. It achieves a stability ARI of 0.887, an external-label ARI of 0.479, a purity of 80.5%, and a composite score of 0.470. Its relatively strong performance shows that substantial role information exists outside direct combat and weapon specialization. Although this model performs well, it combines several feature groups and therefore does not demonstrate that any individual feature category is sufficient.

Removing weapon-share features increases external ARI to 0.512 and purity to 81.3%, while removing combat features produces an ARI of 0.509 and the same 81.3% purity. These improvements suggest that weapon specialization and individual combat performance can sometimes dominate the clustering, pulling players toward mechanical or performance-based groups rather than the broader tactical categories represented by the external labels. Although these ablations improve agreement with the external labels, they should not automatically be considered superior to the full model. The external annotations represent one expert-defined role framework, while the full model may preserve behavioral distinctions that are not captured by those labels.

Trading and teamwork features have limited influence on the final T-side solution. Removing them produces an external ARI of 0.480, purity of 79.7%, and stability of 0.833, all comparable to or slightly better than the full model. Trading and teamwork features alone perform poorly against the external labels, producing an ARI of 0.113. These results do not support treating trading as the primary driver of the T-side role structure.

Removing positioning and movement features reduces external ARI to 0.431, purity to 77.2%, and the composite score to 0.382. The decline is meaningful but smaller than the corresponding CT-side decline. T-side roles therefore depend on spatial behavior, but they are not defined by positioning alone. Contact timing, opening aggression, combat behavior, weapon usage, and utility provide additional signals that preserve part of the structure when positional features are removed.

The category-only models demonstrate the difference between internal clustering quality and external role agreement. Combat-only features achieve the highest T-side composite score of 0.492 and a silhouette score of 0.283, but their external ARI is only 0.268. Utility-only features similarly produce relatively compact clusters but achieve an external ARI of only 0.274. Positioning and movement features alone produce an ARI of 0.262. These categories produce separated groups among players, but none independently reproduces the full expert-defined role structure.

Overall, the T-side ablations suggest that role structure is distributed across several interacting feature groups. Weapon and combat variables create strong separation but can compete with broader tactical alignment. Positioning remains important, particularly for identifying Lurkers, while utility and contact-related features help characterize the lower-output, IGL-enriched Spacetaker group. The T-side results are therefore more overlapping and context-dependent than the CT-side results, supporting the interpretation that attacking responsibilities are expressed through combinations of behavior rather than through one dominant feature category.

Limitations

The most significant limitation is the sample size. Although the dataset contains 463 professional demos, the final clustering dataset only covers 214 players, which is small for unsupervised learning. This is amplified by the narrow player pool, which consists of top-tier Counter-Strike professionals, creating a compressed skill distribution where meaningful role differences may be subtle and overlapping.

The external role labels should also be interpreted with caution. This project uses expert curated role annotations from a database maintained by Harry “NER0cs” Richards, a Counter Strike analyst and

writer for HLTV.org. These labels provide a strong reference point, but they should still be treated as expert labels rather than absolute ground truth. Professional CS2 roles vary by map, side, opponent, economy state, roster context, and team system. Some players also occupy multiple roles simultaneously, such as an AWPPer who also serves as an in-game leader. As a result, disagreement between clusters and external labels may reflect role overlap or hybrid responsibilities rather than model failure.

Another limitation is that the current feature set averages behavior across maps. This produces more stable player level profiles, but it removes important context. A player may anchor a bombsite on one map, rotate actively on another, and take different T-side responsibilities depending on the team's playbook. Map stratified clustering would likely recover more specific role patterns, especially on the CT-side, but would require a much larger sample size.

The clustering results are also constrained by the available feature representation. The model has no access to voice communication, mid-round calls, set strategies, opponent specific preparation, or intended tactical responsibilities. This is particularly important for the IGL heavy T-side cluster. The model can only detect indirect behavioral traces of calling, such as utility usage, teammate proximity, and contact patterns. Therefore, this cluster should be interpreted as evidence that some T-side calling responsibilities may produce measurable behavioral side effects, not as proof that in-game leadership is directly observable from demo statistics alone.

Conclusion

This project shows that professional CS2 player roles are only partially recoverable from aggregated behavioral and positional gameplay data. Across model families and side specific datasets, AWPers formed the clearest and most stable clusters, indicating that weapon profile features create a much stronger role signal than most rifler behaviors. The model also recovered meaningful non AWPPer structure, including a CT-side Rotator/Anchor split and a T-side Lurker group, but these clusters were less pure.

The T-side IGL result is better understood as a role overlap finding than as a clean IGL discovery. Its profile suggests a lower resource, utility oriented space taking role, with stronger utility usage and closer team spacing. This matches the way many modern IGLs are used on the T-side, where they create space and enable higher firepower teammates.

Because the model only observes gameplay behavior, this cluster cannot determine who is calling or making strategic decisions. The result should therefore be interpreted narrowly. Some T-side IGL

responsibilities appear to leave measurable traces in utility usage, spacing, and role overlap, but IGLs are not cleanly separable as a global category from demo statistics alone.

Overall, the results suggest that CS2 roles are not equally discoverable from aggregate player statistics. AWPers are highly separable, some positional rifler roles are moderately recoverable, and broader rifler or IGL categories remain noisy because they vary by map, team system, and tactical context. The most valuable extension would be map-stratified analysis, which could preserve positional structure that is currently averaged away across maps.

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